

Modeling relationship between Stroke and Categorical Factors

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Abstract:

This study used a data sample of 5110 individuals consisting of men and women. From these men and women, we obtained information such as age, marital status, work, residence, average glucose levels, smoking history, BMI, heart disease, and whether the individual had suffered a stroke or not in the past. The age of the individuals began from as young as a couple months and as old as 85. For this paper we will be focusing on categorical variables such as marital status, work, residence, smoking history, heart disease, and gender. For each categorical variable a table was crafted to compare the variable next to strokes. Then the standardized residuals and chi-squared test was computed to see which variable needed further investigation. A mosaic plot was made to compare and further show which variables had a greater effect on the chance of getting a stroke. For the case of the age variable a regression model was calculated using the logit function and then was fitted for the age values. We then graphed the regression model against the proportion of individuals who got a stroke at each age value. Variables that were left out of this paper altogether include the BMI and average glucose levels. This study found that marital status, age, heart disease, and hypertension all played a role in the individual experiencing a stroke. A surprise finding was that while the residuals imply that smoking influences whether you get a heart stroke or not the G^2 partition led to some confounding answers. Aside from this all other finding were in align with what is currently known about strokes and what causes them. Other factors such as residence and gender were found to have little to no effect on the individuals and strokes, and so subsequently they were not pursued further.

Introduction:

Within our lifetime 1 in 4 people will have suffered a stroke. Furthermore, every year 790,000 people suffer from strokes and over 100,000 of them will not survive. With such a high mortality rate strokes are a very serious issue, and it would be best to find the main causes of strokes to find ways to prevent it. To solve this question, we will look at a dataset that was obtained from Liril Kumar Amal from Kaggle.com. Within this dataset are some commonly believed factors of strokes and we will compute residuals and relative risk to see which play a role in increasing the chance of stroke and by how much compared to individuals who do not have these factors. We also take the time to study some factors such as gender that intuitively should not play a role in whether one will experience a stroke. The relationship between marriage status and stroke is something that while could play a role in stress would be interesting to see how it compares to the other variables such as hypertension, and age.

Preliminary Analysis:

One of the first things that had to be done was to go over the data set and check for any missing variables or values. For this dataset the only difficult value to work around was that for gender there was one individual who was recorded as neither male or female and went by other. Because of this when the chi-squared test was ran we got the expected values to be

stroke		
gender	No	Yes
Female	2848.108415	145.89158513
Male	2011.940313	103.05968689
other	0.951272	0.04872798

This is bad because for the chi-square test to be valid all expected values must be greater than or equal to 5. But because there is only one of these values, we can leave it out without having that much of an affect on our overall calculations. After removing the other gender and rerunning the chi-square test, we get a p-value of .516 and with $\alpha=.05$ we fail to reject the null hypothesis meaning that what gender you are does not affect whether if you get a stroke or not. This is further backed by the residuals being close to 1 which means that no cell provides strong evidence against null. The residuals for each 2x2 table is shown below.

```
> gender_residuals
stroke
gender      No      Yes
Female  0.6490565 -0.6490565
Male    -0.6490565  0.6490565
> hypertension_residuals
stroke
hypertension No      Yes
no    9.143111 -9.143111
yes   -9.143111  9.143111
> heartdisease_residuals
stroke
heart_disease No      Yes
no    9.644228 -9.644228
yes   -9.644228  9.644228
> marriage_residuals
stroke
ever_married No      Yes
No    7.744587 -7.744587
Yes   -7.744587  7.744587
> workresiduals
stroke
work_type      No      Yes
children  5.9953328 -5.9953328
Govt_job  -0.1913423  0.1913423
Never_worked 1.0638615 -1.0638615
Private   -0.8498217  0.8498217
Self-employed -4.4440521  4.4440521
> residence_residuals
stroke
Residence_type No      Yes
Rural  1.105001 -1.105001
Urban  -1.105001  1.105001
> smoking_residuals
smoking_status
stroke formerly smoked never smoked smokes unknown
No    -4.6147078  0.2951362 -0.6390124  3.9953777
Yes    4.6147078 -0.2951362  0.6390124 -3.9953777
```

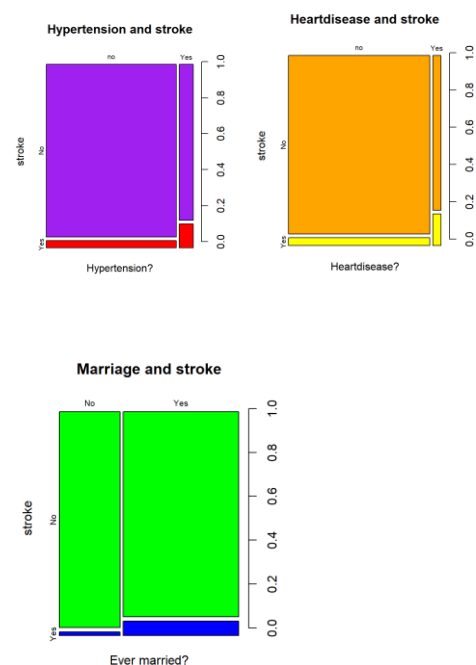
As shown gender and residence has residuals that are close to 1 and so they do not provide evidence against the null meaning that stroke and residence are not related. This leaves hypertension, heart disease, marriage status, and the type of work they do, and smoking history. We then checked to make sure that the expected values for each 2x2 table was greater than 5. So from the remaining only type of work, they do had a cell that had an expected value of less than 5. Because of this we could not run the next step, so we leave it. Now let us take a moment to focus on the smoking history dataset.

From the residuals we see that cells that would provide strong evidence against null are only the unknown and formerly smoked ones. To try to see what was going on I tried to find a way to partition the table. First, I partitioned the table to have formerly smoked and never smoked together then calculating the G^2 got 10.510 when the original G^2 of the nonpartitioned table was 28.110. This means that there is not a big difference between former smokers and people who have never smoke so we cannot combine these cells. Then I tried to compare the never smoked group and the smoker group. The G^2 for this partition was approximately .376 and this would imply that the two tables are not that different from each other, and you could combine them, but this makes no sense as this means that this would mean that smokers and nonsmokers are similar when it comes to their chance of getting a stroke. The two possible ways such a result could arise is from either some other variable is greatly affecting the outcome of smokers or maybe it is in part with the unknown section of the table. The unknown is taking people that should be going into the smoker category which is then making it seem as if the chance of smokers and nonsmokers getting a heart stroke are the same. Because of this we are not able to use the smoking category. So, then we are left with hypertension, marriage history, and heart disease.

Main Analysis:

For marriage history, heart disease, and hypertension I computed the relative risk to see how big of a difference having a

any of these conditions would be compared to not. So, for individuals with hypertension, we get a Relative Risk (RR) of 3.340, then for those who have heart disease we get a $RR=4.075$, and for individuals who were married their RR is 3.975. This means that for all three categories the proportion of individuals that have hypertension, heart disease, and have been married have a higher chance of having a heart stroke. Not only that but for all three categories the proportion is over 200% percent higher for each one. All relative risk and confidence interval calculations are shown on the table on the next page. To visualize this data, I made a mosaic graph for each one below.

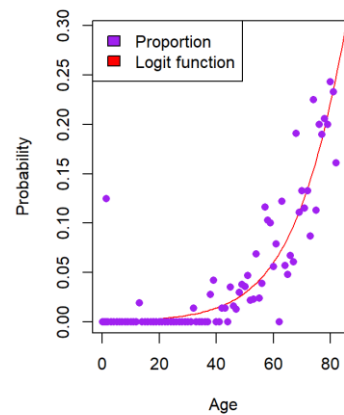


As we can see for each case the proportion of individuals that got a stroke given that they had hypertension or heart disease or were married is much higher than the proportion of people who had no complication but still got a stroke. The final category that we are going to consider is age. Unlike the previous relative risk would not be a good choice. Instead, we will use a generalized linear model. Because the outcome is yes or no for the random component, we will use a binomial distribution with a logit link function. After the computation we obtain a $\text{logit}(\mu) = -7.231 + .0747\text{Age}$.

Furthermore, because we get a $p\text{-value} < 2e^{-16}$ and using $\alpha = .05$ we reject the null hypothesis and say that there is a relationship between your age and the chance of getting a heart stroke. From the equation we can say that there is a positive correlation between your age and chance of getting a stroke as there is a positive slope. We say that for 1 unit increase in age the odds of a stroke multiply by 1.078. The graph on the right was created by first calculating the proportion of individuals who had a stroke given their age (from dataset) and then plotting them with the purple dots. Then the red line is the logit function where we fitted it with the values from the dataset. As we can see the graph does an alright job of estimating the proportion from the data set.

Without a doubt the variables Age, hypertension, Marriage status, and heart

disease all play a role in increasing the chance of a stroke. Furthermore, from what was shown in the



data the variables of gender and residence did not affect the chance of a stroke significantly, given our alpha level of .05. From the Relative Risk we saw that heart disease increased the proportion of individuals who have a stroke the most. Then it was followed by Marriage status and last place was whether someone had hypertension.

Further analysis can be done to see what is affecting the smoking behavior and why based on the residuals the only cells that have a high chance of rejecting the null hypothesis is the former smokers and the smokers that are unsure of their smoking habits. Something that might have led to this occurrence may come from setting up a 2x2x2 table with smoking habits as one factor, average glucose levels as another, and stroke as the final one.

Table	RR (95% C.I)
Hypertension and Stroke	3.340049(2.556093,4.341896)
Heart Disease and Stroke	4.075154(3.027318,5.427984)
Marriage status and Stroke	3.975236(2.719727,5.820717)

References

1. "Heart Stroke." *Kaggle*, 26 Oct. 2020, www.kaggle.com/datasets/lirilukumaramal/heart-stroke.
2. Agresti, Alan. *An Introduction to Categorical Data Analysis*. Wiley-Interscience, 2007.